Five-Year Chronosequence of Unassisted Natural Regeneration in a Premontane Forest Fragment, Costa Rica: Microclimatic Stability and Rare Taxa Recruitment (2021–2026)

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Piedades Sur, San Ramón, Alajuela, Costa Rica (10.104583, -84.564639) | 1,250 m.a.s.l.

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Abstract

We present a five-year dataset (Feb 2021–Feb 2026) documenting Unassisted Natural Regeneration (UNR) in a 5,800 m² premontane forest reserve in Piedades Sur, Costa Rica (1,250 m.a.s.l.). The site contains a patch of 4,000 m² primary forest core surrounded by an active regeneration zone that was used for low pasture in 2021. A floristic baseline of 26 woody species was established in 2021; current monitoring documents natural recruitment (seedlings and shrubs ≤2 m height) in a 30×30 m plot within the regeneration zone. Of the 26 baseline species, 54% are classified as rare or uncommon. The dataset is complemented by four years of high-resolution microclimatic monitoring (2022–2025), revealing exceptional thermal homeostasis ($\sigma \sim 0.2$°C interannual variance). A historic polar air intrusion in February 2026 provided natural validation of the site's thermal buffering capacity. All observations are georeferenced and publicly validated via iNaturalist (*Project: Reserva Huaravito / Sitio Natural Sagrado*; 129 observations, 46 species) and cross-referenced with Terraware (Terraformation App), Restor.eco, and FERM. This dataset directly addresses SNAPP-MORE's objectives of identifying cost-effective, early monitoring indicators that predict long-term restoration trajectories in tropical forest systems.

1. Study Area

The Huaravito Reserve Sacred Natural Site (5,800 m² total; 4,000 m² primary forest core) is located in Quebrada Honda, Quebradillas, Piedades Sur, San Ramón, Alajuela Province, Costa Rica (10°06'16.5"N 84°33'52.7"W) at 1,250 m.a.s.l. The site is classified as premontane humid forest (Holdridge Life Zone system). The primary forest core is surrounded by a transition zone (buffer) that was active cattle pasture until 2021, when livestock were excluded and passive regeneration was initiated. The surrounding landscape matrix is mixed agriculture and secondary growth, with the reserve positioned as an isolated forest fragment along a stream corridor (Quebrada Honda - Quebradillas). Management follows the Open Standards for the Practice of Conservation, formalized in a General Management Plan (2021–2030).

2. Dataset Structure & Methods

2.1 Floristic Baseline (Year 0, 2021)

A comprehensive floristic inventory of the primary forest core was conducted in 2021, documenting 26 woody species across 19 plant families. For each species, the following attributes were recorded: scientific name, common name (including indigenous Cabécar and Bribí nomenclature where available), plant family, growth form (tree/shrub), IUCN Red List status, and rarity classification (common/uncommon). This baseline serves as the "memory" of the system and as the control group for measuring natural recruitment success.

2.2 Natural Recruitment Monitoring (Year 1–5, 2021–2026)

Ongoing monitoring focuses on natural recruits (seedlings and woody shrubs ≤2 m total height) in a 30×30 m plot within the former pasture zone adjacent to the primary forest core. Monitoring was conducted using Terraware for standardized data entry and GPS georeferencing, and NASA GLOBE Observer (Trees module) for height measurement and canopy cover estimation compatible with ICESat-2/GEDI orbital validation. All species observations are independently validated via iNaturalist (Project: Reserva Huaravito / Sitio Natural Sagrado; URL: https://www.inaturalist.org/projects/reserva-huaravito-sitio-natural-sagrado), providing public photographic vouchers, timestamps, GPS coordinates, and community-based taxonomic verification.

For each recruit, the following Key Ecological Attributes (KEAs) are recorded:

- Vegetative vigor: herbivory score (0–4 scale) and foliage color
- Microsite conditions: canopy cover class (open/partial/closed)
- Dispersal connectivity: distance to nearest mature conspecific or large-canopy tree (m)
- Structural development: total height (m) and basal diameter (cm)

2.3 Microclimatic Monitoring (2022–2025)

A digital temperature sensor was deployed within the primary forest core at a fixed height of 1.5 m, recording continuous temperature data from 2022 to present. Data was aggregated to monthly means and cross-validated against regional weather station data (San Ramón area). Microclimatic decoupling was calculated as the differential between internal forest temperature and external ambient temperature.

3. Dataset Summary

3.1 Floristic Baseline: 26 Core Species

Scientific Name	Family	Growth Form	Successional Guild	Rarity	IUCN
Persea schiedeana	Lauraceae	Tree	Late-successional	Common	**EN**
Quetzalia occidentalis	Celastraceae	Shrub	Late-successional	Uncommon	—
Ruprechtia costata	Polygonaceae	Shrub	Late-successional	Uncommon	—
Mespilodaphne veraguensis	Lauraceae	Tree	Late-successional	Common	—
Roupala montana	Proteaceae	Tree	Late-successional	Common	—
Myrsine coriacea	Primulaceae	Tree	Late-successional	Uncommon	—
Inga acuminata	Fabaceae	Shrub	Early-intermediate	Uncommon	—
Trichilia martiana	Meliaceae	Tree	Early-intermediate	Common	—
Myrcia splendens	Myrtaceae	Tree	Early-intermediate	Uncommon	—

Scientific Name	Family	Growth Form	Successional Guild	Rarity	IUCN
Myrciaria floribunda	Myrtaceae	Shrub	Early-intermediate	Common	—
Siphoneugena dussii	Myrtaceae	Shrub	Early-intermediate	Uncommon	—
Psidium friedrichsthalianum	Myrtaceae	Tree	Early-intermediate	Common	—
Psidium guajava	Myrtaceae	Tree	Pioneer	Common	—
Vismia baccifera	Hypericaceae	Tree	Pioneer	Common	—
Miconia xalapensis	Melastomataceae	Tree	Pioneer	Common	—
Zanthoxylum rhoifolium	Rutaceae	Tree	Pioneer	Uncommon	—
Clethra mexicana	Clethraceae	Tree	Early-intermediate	Uncommon	—
Calycophyllum candidissimum	Rubiaceae	Tree	Early-intermediate	Common	—
Hauya elegans	Onagraceae	Tree	Early-intermediate	Uncommon	—
Syzygium jambos	Myrtaceae	Tree	Early-intermediate	Uncommon	—
Aphelandra dolichantha	Acanthaceae	Shrub	Early-intermediate	Uncommon	—
Doliocarpus dentatus	Dilleniaceae	Shrub	Early-intermediate	Uncommon	—
Erythroxylum havanense	Erythroxylaceae	Shrub	Pioneer	Common	—
Machaerium biovulatum	Fabaceae	Shrub	Early-intermediate	Uncommon	—
Philodendron tripartitum	Araceae	Shrub	Early-intermediate	Uncommon	—
Xylosma hispidula	Salicaceae	Shrub	Early-intermediate	Common	—

Table 1. Floristic baseline inventory (2021). Successional guild classifications based on shade tolerance, seed size, and growth rate. EN = Endangered (IUCN). — = not listed.

3.2 Year 5 Natural Recruitment: East Edge Regeneration Zone (February 17, 2026)

Sampling conditions: moderate wind (Trade Wind influence), clear sky, WiFi-active field session. Land use history: low-intensity livestock until 2021, full exclusion thereafter. Matrix pressure: agricultural road, abandoned pastures, residual grasses. Sampling type: opportunistic observational confirmation along east edge transect. All 7 taxa uploaded to iNaturalist on collection date.

Key site-level finding: Elephant grass (Pennisetum purpureum) cover estimated at ~1% of plot area, indicating successful suppression of grass dominance within 4–5 years of livestock exclusion — a critical bottleneck in tropical regeneration that commonly stalls succession for >10 years in comparable systems.

ID	Species / Taxon	Guild	Dispersal Mode	Height Class	Density	Grass Cover	Recruitment Signal
001	Candelero sp.	Early–Mid	Zoochorous	**>1.5 m**	~3 ind/10 m²	~1%	Established juvenile
002	*Cuphea appendiculata*	Pioneer / Edge	Generalist	**>1.5 m**	~3 ind/10 m²	~1%	Transitional mosaic
003	Myrtoideae sp.	Mid–Late	Zoochorous	**>1.5 m**	~3 ind/10 m²	~1%	Shade-tolerant recruit
004	Ficus sp. (Moraceae)	Mid	Zoochorous	**>1.5 m**	~3 ind/10 m²	~1%	Faunal dispersal signal
005	Aguacatillo (Lauraceae)	**Late**	Zoochorous (avian)	**>1.5 m**	~3 ind/10 m²	~1%	Core propagule export
006	Maquilla sp.	Mid	Zoochorous	**>1.5 m**	~3 ind/10 m²	~1%	Understory establishment
007	Annona sp.	Early–Mid	Zoochorous	**>1.5 m**	~3 ind/10 m²	~1%	Edge recruitment

Table 2. Recruitment Dynamics Table — East Edge Regeneration Zone, February 17, 2026. All 7 taxa recorded at >1.5 m height class, indicating establishment with minimum 2–3 years of survival (not recent germination). Bold guild/height entries indicate late-successional taxa. Density estimated visually (~3 ind/10 m²); not extrapolated as exact plot count. Elephant grass cover ~1% confirms successful suppression of the primary competitive bottleneck in tropical pasture regeneration. 86% of recorded taxa are zoochorous, confirming functional fauna-flora dispersal network. All records validated in iNaturalist on collection date.

Site-level synthesis: Within 4–5 years of livestock exclusion, the east edge regeneration zone exhibits moderate-density recruitment (>1.5 m height class) of zoochorous mid- and late-successional taxa, with minimal elephant grass persistence (~1% cover), indicating successful suppression of grass dominance and a positive successional trajectory under high matrix pressure.

3.3 iNaturalist Project Summary (as of February 17, 2026)

Metric	Value	Taxonomic Scope	URL
Total observations	**45**	Plants, fungi, fauna	inaturalist.org/projects/reserva-huaravito-sitio-natural-sagrado
Total species documented	**36**	Incl. orchids, bryophytes, arthropods	
Temporal range	May 2021 – Feb 2026	5-year span	
Terraware cross-reference	Apr 2021	Woody species only	terraware.io/species?organizationId=213

Table 3. iNaturalist project statistics. Notable records include two epiphytic orchid species (Scaphyglottis micrantha, S. crurigera) and bryophyte/lichen records indicative of primary forest microclimate conditions.

4. Microclimatic Dataset (2022–2025)

4.1 Four-Year Thermal Summary

Parameter	2022	2023	2024	2025	Mean
Dry season mean (Dec–Apr)	19.9°C	20.1°C	19.8°C	20.0°C	**20.0°C**
Wet season mean (May–Nov)	19.2°C	19.4°C	19.1°C	19.3°C	**19.3°C**
Annual mean	**19.4°C**	**19.6°C**	**19.5°C**	**19.5°C**	**19.5°C**
Interannual variance (σ)	**~0.2°C (exceptional thermal homeostasis)**				

Table 4. Multi-year thermal matrix, Toyopán forest core, 2022–2025. Annual range: 2.1°C (18.8°C min – 20.9°C max). Biogenic cooling effect: -3.7°C relative to San Ramón regional ambient temperature. Interannual variance of σ ~0.2°C is characteristic of intact primary cloud forest.

4.2 Extreme Event Validation: Mesoamerican Polar Air Intrusion (Jan 30 – Feb 9, 2026)

A historic polar air mass intrusion affecting the Mesoamerican isthmus in late January and early February 2026 (documented snowfall in Guatemalan highlands; ice formation on Costa Rican volcanic range) provided a natural stress test for the site's thermal buffering capacity:

- Regional ambient (San José, 1,100 m.a.s.l.): 12.2°C (30-year minimum record)
- External exposure (Toyopán site, 1,250 m.a.s.l.): 13.0°C with 90 km/h wind gusts (est. wind-chill: 10.0°C)
- **Forest core interior: maintained 16.2°C minimum**
- Total thermal buffering capacity: +4.0 to +6.0°C; full operating range during event: 4.9°C (16.2°C to 21.1°C)

This buffering prevented cold-stress mortality in late-successional taxa and sensitive soil microbiomes during a 1-in-100-year climatic event, providing empirical evidence of active microclimate regulation at the site.

5. Key Preliminary Findings

- Successional mosaic confirmed: pioneer (19%), early-intermediate (58%), and late-successional species (23%) co-occur within the same 30×30 m plot after 5 years of passive regeneration, contrary to a simple linear succession model.
- Late-successional species at significant height: Quetzalia occidentalis and Lauraceae recruits (IDs 001 and 005) recorded at ~1.5 m in former pasture, 5 years post-protection.
- 54% rare/uncommon species in baseline, with active recruitment confirmed for specialist taxa including Quetzalia occidentalis (rare shrub) and Persea/Nectandra (large-seeded, mammal-dispersed Lauraceae).
- Ficus sp. (Moraceae) and Annona sp. document spontaneous colonization beyond the 26-species baseline, indicating active zoochorous dispersal by birds and/or bats.
- Epiphytic orchids (Scaphyglottis micrantha, S. crurigera) recorded on host trees confirm primary forest microhabitat conditions within the core zone.
- Thermal stability (σ ~0.2°C interannual variance) is proposed as a high signal-to-cost monitoring indicator for restoration success in tropical montane systems.

6. Limitations

The authors acknowledge the following limitations of the current dataset. The monitoring plot (30×30 m) represents a single spatial unit; additional replicate plots would strengthen spatial inference. Individual plant counts per species in the recruitment census are available but not yet systematically aggregated across all years. Microclimatic data derives from a single sensor; a distributed sensor network would provide spatial resolution of the thermal gradient between core and matrix. Causal attribution of thermal stability to rare taxa recruitment remains correlational pending formal statistical analysis with comparative sites.

7. Data Availability

The dataset is available for sharing with SNAPP-MORE under a standard Data Use Agreement. All primary observations are publicly accessible via the platforms listed below. The authors request: anonymization of site-specific results in publications; generalized location data (regional level) in any public datasets; and the opportunity to review manuscripts prior to submission.

Platform	URL / Identifier	Data Type
iNaturalist	inaturalist.org/projects/reserva-huaravito-sitio-natural-sagrado	45 obs., 36 spp., 2021–2026
Terraware	terraware.io/species?organizationId=213	Woody species inventory
Restor	restor.eco/es/sites/c4c9b69f-1f0b-40bc-9d80-2d18fd1e54cf	Site profile & restoration data
FERM (FAO)	ferm.fao.org/search/initiatives/D0D3iHkwpYYz592W7ooP	Global reporting
Raw CSV (floristic baseline)	Available upon request	26 species, all attributes
Microclimatic dataset	Available upon request	Monthly means, 2022–2025

8. Contact & Collaboration

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